

Likelihood:

$$y_{i,t} \sim \mathcal{N}(\mu_{i,t} + \epsilon_{i,t}, \tau_i), \quad t > 1$$

Where y is the raw weekly rate, i is the health region, and t is week

Mean structure:

$$\begin{aligned} \mu_{i,t} = & \alpha_i && \leftarrow \text{Baseline} \\ & + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) && \leftarrow \text{Annual cycle} \\ & + \delta_{i,t} \mathbf{1}(t = \text{New Year week} - 2) + \delta_{i,t} \mathbf{1}(t = \text{New Year week} - 1) && \leftarrow \text{New Year effect} \\ & && + \delta_{i,t} \mathbf{1}(t = \text{New Year week}) \\ & && + \delta_{i,t} \mathbf{1}(t = \text{New Year week} + 1) \\ & + \kappa_{i=\text{MW},t} \mathbf{1}(t = \text{Aug 8 2024 week}) + \kappa_{i=\text{MW},t} \mathbf{1}(t = \text{Aug 15 2024 week}) && \leftarrow \text{Mid West reset} \\ & && + \kappa_{i=\text{MW},t} \mathbf{1}(t = \text{Aug 22 2024 week}) \\ & && + \kappa_{i=\text{MW},t} \mathbf{1}(t = \text{Aug 29 2024 week}) \\ & && + \kappa_{i=\text{MW},t} \mathbf{1}(t = \text{Sep 5 2024 week}) \end{aligned}$$

AR(1) Component:

$$\epsilon_{i,t} = \phi(y_{i,t-1} - \mu_{i,t-1})$$

Uninformative priors:

$$\alpha_i, \beta_i, \gamma_i, \delta_{i,t}, \kappa_{i=\text{MW},t} \sim \mathcal{N}(0, 0.001)$$

$$\tau_i \sim \Gamma(0.001, 0.001) \quad \phi \sim \text{Uniform}(-1, 1)$$

Normal priors $\mathcal{N}(\mu, \tau)$ use precision in the 2nd argument.